

# Ken (Tongli) Su

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CS PhD student at Emory working on **LLM agents, alignment, interpretability, and safety**, especially multi-agent collaboration and theory of mind: agents that reason about collaborators' beliefs, adapt to human values, and know when to seek human input. Co-first author of LARGER, a retrieval method for LLM coding agents that outperforms seven baselines including Claude Code and Codex; first author on an EMBC 2026 oral. Reviewer, NeurIPS 2026.

## Education

### Emory University

Ph.D. in Computer Science

Atlanta, GA

Aug 2025 – Dec 2028 (expected)

### Columbia University

M.S. in Data Science, GPA: 3.95/4.00

New York, NY

Dec 2023

### Emory University

B.A. in Computer Science & B.A. in Economics (Dual Major), GPA: 3.84/4.00

Atlanta, GA

May 2022

## Selected Publications & Research

- [1] Y. Hu\*, T. Su\*, L. Zhao, B. Zhu, H. Haque. "LARGER: Lexically Anchored Repository Graph Exploration and Retrieval." *arXiv:2605.16352*, 2026. (Under review; \*equal contribution.)
- Formalized **Lexically Anchored Structural Localization**: anchor a coding agent's own lexical search hits into a repository graph, then expand confidence-filtered neighborhoods *inside* its existing search loop — delivering structural evidence (imports, call chains, code–test links) with no external graph database, graph tools, or added agent actions.
  - Built the system end to end: multi-language **AST-based graph construction**, edge weighting, and community detection for graph quality; sidecar storage, dynamic lexical propagation, and budget-aware filtering for retrieval efficiency.
  - +**13.9 Acc@5** (tuned), +**11.8** (fixed hyperparameters), and +**12.2 Recall@5** over the strongest of **7 baselines** (BM25, Agentless, LocAgent, SWE-agent, OpenHands, Claude Code, Codex), with consistent gains across **4 benchmarks** spanning localization, test generation, and codebase QA.
- [2] T. Su, A. Rafiei, M. van Assen, R. Sameni, G. D. Clifford, F. Marzbanrad, N. Katebi. "Cross-Modal Generative Framework for Signal Translation from Fetal–Maternal Electrocardiograms to Fetal Doppler Waveforms." *IEEE EMBC 2026*. **Accepted — Oral Presentation**.
- Designed a cross-modal generative architecture (dilated convolutions + cross-modal attention + self-attention) synthesizing fetal Doppler envelopes from dual-lead ECG, trained on **885 synchronized segments from 39 pregnancies**: **PSD MSE 49.9 dB<sup>2</sup>** (51% **below** the two-channel baseline) and **4.71 bpm** heart-rate error.
  - Showed via ablation that naive maternal-ECG concatenation yields no benefit while **cross-modal attention cuts PSD MSE 39%** — quantifying maternal–fetal cardiac coupling as selective and time-varying.
- [3] T. Su, N. Barari, N. Katebi. "A Review of Multimodal Pretraining Strategies for Physiological Wearable Sensor Foundation Models." *Under review*, 2026. — Synthesized **22 papers (2024–2026)** into a taxonomy of five pretraining paradigms (contrastive, generative, language-grounded, missing-modality, cross-modal distillation) across PPG/ECG/EEG/ACC.

## Industry Experience

### Machine Learning Engineer, GenAI | DuPont Specialty Products USA, LLC

New York, NY | Jul 2024 – Aug 2025

- Built a scalable lead identification and prioritization pipeline on **Azure Databricks** (Spark, Delta Lake, generative models), powering an internal co-pilot that became the company's primary method for identifying prospects — **\$60M+** estimated business value.
- Fine-tuned **Sentence-BERT** with triplet loss for retrieval and ranking: **Recall@5 0.86**, **MRR 0.64** (+27% **over the incumbent GPT-based system**) with hallucinations eliminated; drove marketing converting to **\$3M** revenue.
- Engineered an **OpenAI API** enrichment module (round-robin load balancing, async processing, batch handling): runtime 10 h → 6 h (40%), token costs **–\$12K/yr**, **60 h** development time saved.

### Machine Learning Engineer | ICBC Financial Service, LLC

New York, NY | Feb 2024 – May 2024

- Developed an in-house **transformer foundation model** with a Bing Search API continual re-pretraining loop; **few-shot** financial report translation reached **BLEU 0.59** (cumulative 4-gram), cutting translation costs **\$30K/yr** and turnaround **50%**.
- Shipped a **Flask** app with embedded parameterized T-SQL: data retrieval time **–40%**, SQL injection risk eliminated, **85% adoption** across 50 team members in month one.

### Machine Learning Scientist | Unilever United States, Inc

New York, NY | Sep 2023 – Dec 2023

- Built a **multi-modal neural network** (PyTorch/Keras) fusing tabular product features with fine-tuned VGG16 image embeddings, improving sales-volume **R<sup>2</sup> by 54%** over an XGBoost baseline and optimizing **\$3M** in supply chain cost.
- Engineered a **PySpark** ETL pipeline over ~100 GB of heterogeneous merchant data; deployed end to end on **GCP** for real-time forecasting, improving operational efficiency **20%**.

## Technical Skills

### LLM & Generative AI

Transformers (Hugging Face), LLM & coding agents, retrieval/ranking (RAG, BM25, graph retrieval), fine-tuning (LoRA, triplet loss), Sentence-BERT, few-shot learning, OpenAI API, evaluation (Acc@k, Recall@k, MRR, BLEU)

### ML & Deep Learning

PyTorch, TensorFlow, Keras, Scikit-learn, LightGBM, XGBoost, Optuna; attention, cross-modal & generative models, self-supervised learning, foundation model pretraining, time-series & biosignals

### Languages & Data

Python, SQL, R, Java, JavaScript, C++, MATLAB, Bash; Spark, PySpark, Dask, Pandas, Databricks, Delta Lake, Airflow, Hadoop, MySQL, PostgreSQL, MongoDB, Flask

### Cloud

AWS (EC2, S3, Redshift), Azure (VM, DevOps), GCP (VM, Cloud Storage, BigQuery); Tableau, Power BI